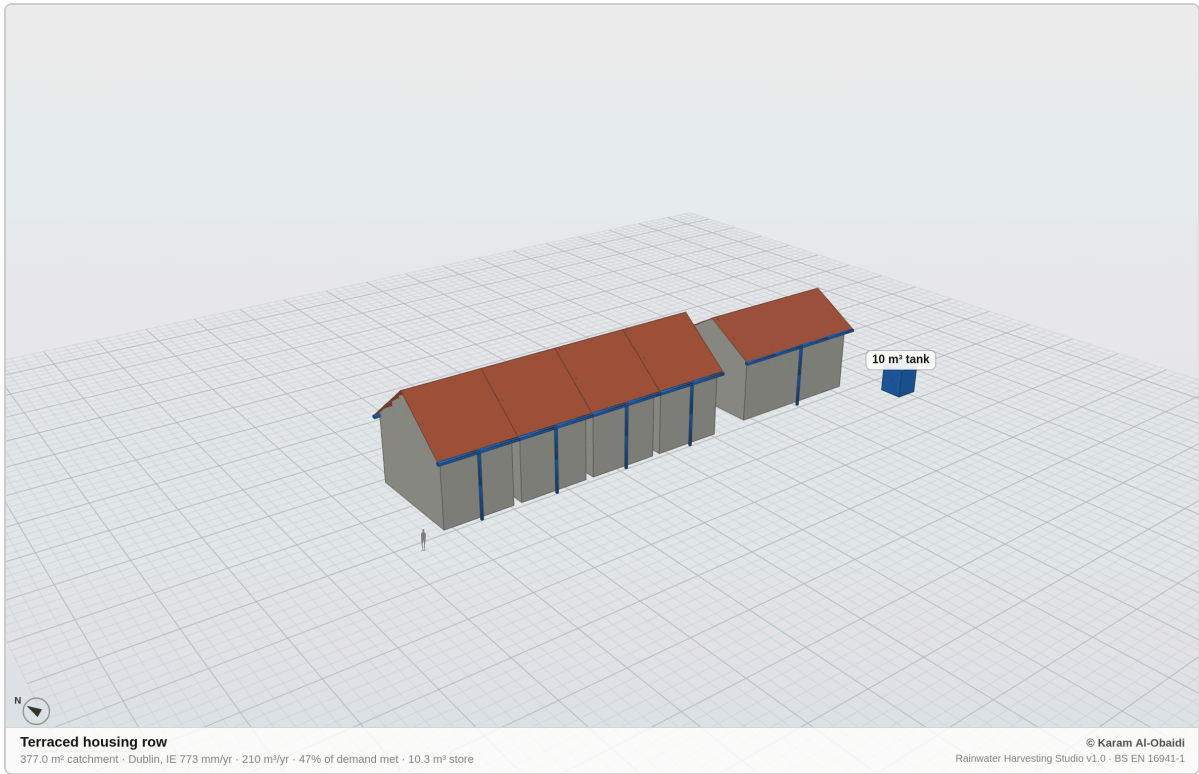


Terraced housing row

Rainwater harvesting appraisal · projected catchment method · BS EN 16941-1:2024



ROOF SCHEDULE

Block	Roof	W×D×H m	Material	Yc	Catch. m²	Surf. m²
House 2	Gable	6.0×9.0×6.0	Pitched roof, rough (concrete tile)	0.80	65.0	82.5
House 3	Gable	6.0×9.0×6.0	Pitched roof, rough (concrete tile)	0.80	65.0	82.5
House 4	Gable	6.0×9.0×6.0	Pitched roof, rough (concrete tile)	0.80	65.0	82.5
House 5	Gable	6.0×9.0×6.0	Pitched roof, rough (concrete tile)	0.80	65.0	82.5
Block 6	Gable	12.0×8.0×6.0	Pitched roof, rough (concrete tile)	0.80	117.0	135.1
Total				0.80	377.0	465.0

MONTHLY FIGURES

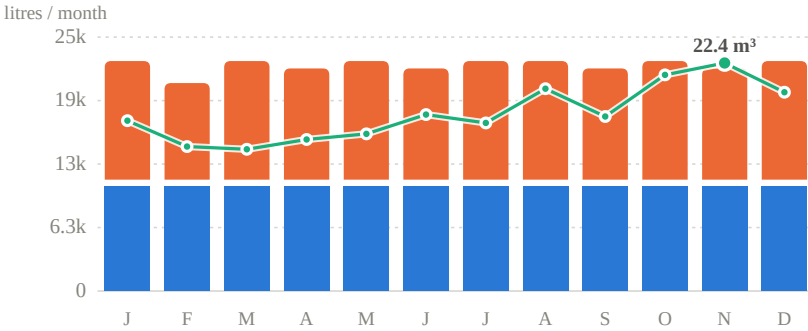
Month	Rain mm	Inflow L	Demand L	Supplied L	Stored L	Spill L	Mains L
Jan	62	16,773	22,320	10,340	10,340	6,434	11,980
Feb	52	14,222	20,160	10,340	10,340	3,882	9,820
Mar	51	13,951	22,320	10,340	10,340	3,611	11,980
Apr	55	14,928	21,600	10,340	10,340	4,588	11,260
May	57	15,470	22,320	10,340	10,340	5,131	11,980
Jun	64	17,370	21,600	10,340	10,340	7,031	11,260
Jul	61	16,556	22,320	10,340	10,340	6,216	11,980
Aug	73	19,922	22,320	10,340	10,340	9,582	11,980
Sep	63	17,180	21,600	10,340	10,340	6,841	11,260
Oct	78	21,279	22,320	10,340	10,340	10,939	11,980
Nov	83	22,446	21,600	10,340	10,340	12,106	11,260
Dec	72	19,569	22,320	10,340	10,340	9,229	11,980
Year	773	209,665	262,800	124,076	—	85,589	138,724

METHOD

$Y = A \cdot e \cdot h \cdot \eta$
 $377.0 \text{ m}^2 \times 773 \text{ mm} \times 0.80 \times 0.90 = 210 \text{ m}^3/\text{yr}$

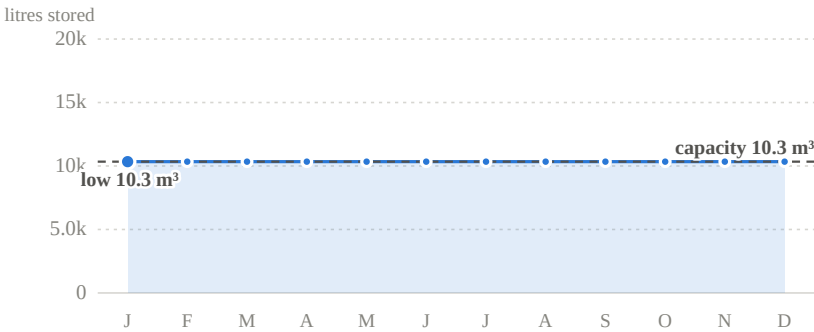
Non-potable demand	263 m³/yr
Supplied from store	124 m³
Mains top-up	139 m³
Overflow to drain	85.6 m³
Water saving efficiency	47%
Coverage rate Cr (A.5)	47%
Months fully supplied (tool's own measure)	0%
Storage capacity	10.3 m³ (14 days)
BS basic approach A.2.1	10.3 m³
Profile-optimised knee	19.0 m³
Mains displaced	124.1 m³/yr

MONTHLY WATER BALANCE



Blue supplied from store · orange mains top-up · green line roof inflow

STORAGE BEHAVIOUR



CAPACITY OPTIMISATION

